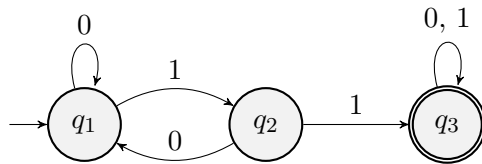


Name: _____

CSSE/MA 474 Worksheet A1

Finite Automata

1. For the finite automaton M_1 , which is the start state? _____ Which is an accept state? _____



What is the output given each input string?

- 01101 → _____
- 00101 → _____

What is the language A of M_1 ?

When A is the language of M_1 , we also say M_1 *recognizes* A and that $A = L(M_1)$.

2. Complete the formal definition: A *finite automaton* is a 5-tuple, $(Q, \Sigma, \delta, q_0, F)$, where...

call me amogus

3. Give a formal description of M_1 above.

4. Strings and languages. Define:

- *string*:
- *language*:
- *empty string* ϵ :
- *empty language* $\emptyset$:

5. Definition. M *accepts* string $w = w_1w_2 \cdots w_n$, where each $w_i \in \Sigma$, if there exists a sequences of states $r_0, r_1, \dots, r_n \in Q$ such that...

6. Definition. M *recognizes* language $L(M) =$ _____

7. Definition. A language is _____ if some finite automaton recognizes it.

8. Give an example of a regular language.

9. Are the following languages regular?

$$B = \{w \mid w \text{ has an even number of 1s}\}$$

$$C = \{w \mid w \text{ has an equal number of 0s and 1s}\}$$